

Programming Assignment #3: Great Circle Distance

1 Overview

In this lab, write a C program to calculate the great-circle distance between two coordinates expressed as latitude and longitude. The program should also calculate the initial and final heading of the great-circle route between the two points. Formula to calculate these quantities and an explanation of their basis available in many places on the web. One very good source of information is [here](#).

Your program should

- Request the user to input the latitude and longitude values for two coordinates.
- The coordinates should be entered in the degrees (integer), minutes (integer), seconds (double), and compass direction (char, i.e., N/S/E/W) format. *You may assume that the user enters the data in the correct format and you do not have to handle errors if he/she does not.*
- You may find it beneficial to use the `math.h` library.
- Degrees are measured in degrees, seconds and minutes. For this program, you will need to convert the users input into a single “degree.” (There are 60 minutes in a degree and 60 seconds in a minute.)
- Remember that South and West are considered negative (down and left).
- You should have some predefined constants. My code uses 7 including the conversions of radians to degrees and degrees to radians.
- Code the appropriate formulas to calculate the great circle distance between two points. Again, you may use the formulas from [here](#). You may use this as a reference for the formulas you need, but you **MAY NOT** use the Java script code provided on the website. You must write your own code using your own variable names.
- Output the distance, initial bearing and final bearing.
- Ask the user if they would like to do another calculation after the current one is completed. Thus the user must be able to do as many sets of calculations as they want until they indicate that they are finished.

2 Code Documentation

The very first lines in *all* of your lab project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, should be the name of the assignment and your name, e.g:

```
printf("Programming Assignment #3 Great Circle Distance - Joe Student\n");
```

The next line of output should be `printf("EGRE 245: Introduction to Programming Using C, Due: March 30, 2025\n");`

3 Sample Run

Your output should match the sample run below.

```
Enter 1st Latitude (DD MM SS.SS N/S): 37 32 43.89 N
Enter 1st Longitude (DD MM SS.SS E/W): 77 26 57.98 W
Enter 2nd Latitude (DD MM SS.SS N/S): 34 08 03.18 N
Enter 2nd Longitude (DD MM SS.SS E/W): 118 19 17.14 W
Distance = 3674.82 km
Initial Bearing = 276.69 degrees
Final Bearing = 252.06 degrees

Do another? Enter 1 for Yes or 2 for No: 1

Enter 1st Latitude (DD MM SS.SS N/S): 37 32 43.89 N
Enter 1st Longitude (DD MM SS.SS E/W): 77 26 57.98 W
Enter 2nd Latitude (DD MM SS.SS N/S): 33 51 25.07 S
Enter 2nd Longitude (DD MM SS.SS E/W): 151 12 55.00 E
Distance = 15650.46 km
Initial Bearing = 260.22 degrees
Final Bearing = 250.20 degrees

Do another? Enter 1 for Yes or 2 for No: 2
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Programming Assignment#3_FirstName_LastName.c`

Due date: The assignment is due at the 11:59 pm on Sunday, November 2nd.

5 Academic Integrity

Any work submitted should be the student’s original work. Submitting code written by AI using topics not covered in class is strictly prohibited. Read the VCU Academic Misconduct standard [here](#).

Be sure to save a copy of your .c file.